

ADAPTION OF FPGA ACCELERATION FOR REAL-TIME SIGNAL PROCESSING

A PROJECT REPORT

Submitted by

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in partial fulfillment of the requirement

for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



CHENNAI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)



MARCH - 2026

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To be an eminent centre for Academia, Industry and Research by imparting knowledge, relevant practices and inculcating human values to address global challenges through novelty and sustainability.

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- IM2.** To transform lives through deployment of emerging technology, novelty and sustainability.
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- IM4.** To contribute towards the research ecosystem by providing a suitable,

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Mission of the Department:

- DM1.** To provide strong fundamentals and technical skills for Computer Science applications through effective teaching learning methodologies.
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BONAFIDE CERTIFICATE

Certified that this project report titled “**ADAPTION OF FPGA ACCELERATION FOR REAL-TIME SIGNAL PROCESSING**” is the Bonafide work of **SRI VISHNU R (24CS0929), SRI GIRI SUDHAN M D (24CS0926)** who carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION

We jointly declare that the project report on “**ADOPTION OF FPGA ACCELERATION FOR REAL-TIME SIGNAL PROCESSING**” is the result of original work done by us and best of our knowledge, similar work has not been submitted to “**ANNA UNIVERSITY CHENNAI**” for the requirement of Degree of **BACHELOR OF ENGINEERING**. This project report is submitted on the partial fulfilment of the requirement of the award of Degree of **COMPUTER SCIENCE AND ENGINEERING**.

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Date:

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ABSTRACT

Real-time signal processing is essential in modern applications such as wireless communication, medical systems, and multimedia technologies, where immediate data analysis is required for accurate and timely responses. Traditional processors like CPUs often face limitations in handling such tasks efficiently due to their sequential execution and higher latency. As the volume of signal data generated from sensors and embedded systems continues to increase, there is a growing need for faster and more efficient processing methods. This project focuses on the use of Field Programmable Gate Arrays (FPGAs) to improve the performance of real-time signal processing systems. FPGAs are reconfigurable hardware devices that enable parallel execution of operations, resulting in reduced processing time and improved efficiency compared to conventional software-based approaches. In this work, signal processing techniques such as filtering are implemented using a Finite Impulse Response (FIR) filter designed in Verilog HDL. The FPGA-based architecture allows faster computation through hardware acceleration and parallel processing. The system is developed and simulated using tools such as Xilinx Vivado and ModelSim to verify functionality and performance. The proposed approach demonstrates that FPGA-based implementations provide lower latency, higher throughput, and better efficiency for real-time signal processing applications. This makes them suitable for high-speed and time-critical systems.

Keywords: FPGA, FIR Filter, Real-Time Signal Processing, ModelSim, Verilog HDL.